Luigi Roberti

Curriculum Vitæ

Leibniz Universität Hannover
Institut für Angewandte Mathematik
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Personal Information

Place of birth	Brescia (BS), Italy
Citizenship	Italian
Gender	Male
Title	Dr.rer.nat.

Current Employment

01/10/2025 – Present **Leibniz Universität Hannover**, Institut für Angewandte Mathematik.
Wissenschaftlicher Mitarbeiter – Postdoc (Postdoctoral Research Assistant).

Past Employment

01/10/2021 – 31/08/2025 **University of Vienna**, Faculty of Mathematics.
Universitätsassistent Praedoc (Predoctoral University Assistant).
Mentor: Prof. Dr. Adrian Constantin.

01/10/2020 – 30/09/2021 **University of Vienna**, Faculty of Mathematics.
Wissenschaftlicher Projektmitarbeiter (Scientific Project Assistant).
Employment within the project "Equatorial wave-current interactions", supported by the WWTF grant MA16-009.
Principal Investigator: Prof. Dr. Adrian Constantin.

Education

10/2020 – 08/2025 **Dr.rer.nat. in Mathematics**, University of Vienna.
Thesis: *Mathematical aspects of geophysical fluid flows*.
Advisor: Prof. Dr. Adrian Constantin.

10/2018 – 09/2020 **MSc in Mathematics**, University of Vienna.
Thesis: *On the decrease of velocity with depth in periodic water waves*.
Advisor: Prof. Dr. Adrian Constantin.

10/2015 – 08/2018 **BSc in Mathematics**, University of Vienna.
Thesis: *Hardy's inequality.*
Advisor: Prof. Dr. Roland Donninger.

10/2014 – 03/2018 **BSc in Physics**, University of Vienna.
Thesis: *Bestimmung der Volumenverteilung eines Aerosols aus der Messung von dessen Extinktionskoeffizienten.*
Advisor: Prof. Dr. Helmuth Horvath†.

09/2009 – 07/2014 **Diploma di Maturità Scientifica (Science High School Diploma)**, Liceo Scientifico Statale Nicolò Copernico, Brescia, Italy.

Publications

Preprints

[1] (With C. Puntini and E. Stefanescu) On large-scale oceanic wind-drift currents. arXiv:2602.06473, 50 pp.

Accepted articles

[2] (With Q. Ding) Variational formulation for a model describing the Antarctic Circumpolar Current with variable density. *Differ. Integral Equ.* **39**(7–8) (2026), 497–510. doi:10.57262/die039-0708-497 (to appear).

Published articles

[3] (With E. Stefanescu) Global-in-time existence, uniqueness, and stability of solutions to a model of the Antarctic Circumpolar Current. *Discrete Contin. Dyn. Syst.* **49** (2026), 289–310. doi:10.3934/dcds.2025169.

[4] (With B.-V. Matioc and Ch. Walker) Quasilinear parabolic equations with superlinear nonlinearities in critical spaces. *J. Differ. Equ.* **429** (2025), 283–317. doi:10.1016/j.jde.2025.02.039.

[5] (With Q. Ding) Stratified ocean gyres with Stuart-type vortices. *Ann. Mat. Pura Appl.* **203** (2024), 2847–2862. doi:10.1007/s10231-024-01469-5.

[6] (With B.-V. Matioc) Weak and classical solutions to an asymptotic model for atmospheric flows. *J. Differ. Equ.* **367** (2023), 603–624. doi:10.1016/j.jde.2023.05.023.

[7] The surface current of Ekman flows with time-dependent eddy viscosity. *Comm. Pure Appl. Anal.* **21**(7) (2022), 2463–2477. doi:10.3934/cpaa.2022064.

[8] The Ekman spiral for piecewise-constant eddy viscosity. *Appl. Anal.* **101**(15) (2022), 5528–5536. doi:10.1080/00036811.2021.1896709.

[9] Perturbation analysis for the surface deflection angle of Ekman-type flows with variable eddy viscosity. *J. Math. Fluid Mech.* **23**(3) (2021), No. 57. doi:10.1007/s00021-021-00586-y.

[10] On the decrease of velocity with depth in irrotational periodic water waves. *Monatsh. Math.* **193**(3) (2020), 671–682. doi:10.1007/s00605-020-01451-2.

Conferences and Seminars

Talks

2026 Invited talk in working seminar, Hangzhou Normal University.

Invited talk at workshop *Fluid Dynamics and Spectral Theory of Differential Operators*, Nanjing University.

2025 Invited talk at *Bielefeld Analysis Seminar*, Universität Bielefeld.

Talk at *Oberseminar Analysis und Theoretische Physik*, Leibniz Universität Hannover.

Invited talk at workshop *Modelling of Fluid Propagation: Mathematical Theory and Numerical Approximation*, International Centre for Mathematical Meetings (CIEM), Castro Urdiales.

Invited talk at *Conference on Mathematics of Wave Phenomena*, Karlsruhe Institute of Technology.

2024 Invited talk at *Mathematical Sciences Seminar*, University College Cork.

Talk at workshop *Mathematical Theory of Water Waves*, Lund University.

Talk at *PDE Afternoon Seminar*, University of Vienna.

Invited talk at *MCMP Seminar*, University of Vienna.

2023 Talk at *ÖMG Tagung 2023*, TU Graz.

2022 Talk at *DMV-Jahrestagung 2022*, Freie Universität Berlin.

Talk at *Seminar Applied and Computational PDE*, University of Vienna.

2020 Talk at *Seminar: Applied PDE*, University of Vienna.

2019 Talk at *Complex Analysis Seminar*, University of Vienna.

Additional conferences and programmes attended

2025 Workshop *Mathematical Advances in Geophysical Fluid Dynamics*, Mathematisches Forschungsinstitut Oberwolfach.

2024 *EWM-EMS Summer School: Water Waves and Nonlinear Dispersive Equations*, Mittag-Leffler-Institut.

Fluid Flows – Analysis and Modelling. Conference in Honour of Robin S. Johnson's 80th Birthday, University of Vienna

2023 *Aspects of Nonlinear Evolution. Conference in Honour of Joachim Escher's 60th Birthday*, Leibniz Universität Hannover.

Workshop on Nonlinear Dispersive Waves (online attendance), University College Cork.

2021 *Summer School of the Vienna School of Mathematics*, Weißensee.

Teaching

2026S Funktionentheorie für das Lehramt (Lecture on complex analysis for prospective teachers, in German), Leibniz Universität Hannover.

Übung zu Funktionentheorie für das Lehramt (Exercise class to my lecture on complex analysis for prospective teachers, in German), Leibniz Universität Hannover.

2024W Proseminar "Komplexe und Harmonische Analysis" (Exercise class "Complex and Harmonic Analysis", in German), University of Vienna.

2024S Proseminar "Analysis 2" (Exercise class "Analysis 2", in German), University of Vienna.

2023W Proseminar "Analysis und Lineare Algebra 1" (Exercise class "Analysis and Linear Algebra 1", in German), University of Vienna.

2023S Übungen zu "Funktionalanalysis" (Exercise class to "Functional Analysis", in German), University of Vienna.

2022W Übungen zu "Partielle Differentialgleichungen" (Exercise class to "Partial Differential Equations", in German), University of Vienna.

2022S Übungen zur Einführung in das mathematische Arbeiten und Rechenübungen (Exercise class to Introduction to mathematics and calculations, in German), University of Vienna.

(Co-)Supervision

2025 BSc thesis of Stefano Mazzeo, University of Vienna (co-supervised with Dr. Jörg Weber).

Peer Review Activity

Referee for *Appl. Math. Comput.* / *Appl. Math. Lett.* / *J. Differ. Equ.* / *J. Math. Phys.* / *Monats. Math.* / *Nonlinear Anal. Real World Appl.* / *Pure Appl. Geophys.* / *Qual. Theory Dyn. Syst.*

Training

2025 Workshop *Diversity in Practice*, University of Vienna.

Computer Skills and Competence

Operating systems	Working knowledge of Windows and Linux OS.
Programming languages	Working knowledge of Python, Fortran, LaTeX language and some of its text editors (Overleaf, Texmaker, TeXworks).
Software programs	Working knowledge of Matlab, Wolfram Mathematica, Gnuplot.

Platforms

- arXiv
- Google Scholar
- ORCID
- ResearchGate
- Scopus

Languages

Italian	Native speaker.
English	Excellent knowledge and understanding of both written and spoken language.
German	Excellent knowledge and understanding of both written and spoken language.

Hannover, April 2, 2026